

Practical 2 (User & Group Management)

1. Create a user with fix UID. (User: HPC, UID:2026)

```
astha@cdac:~$ useradd -u2026 hpc
useradd: Permission denied.
useradd: cannot lock /etc/passwd; try again later.
astha@cdac:~$ sudo useradd -u2026 hpc
[sudo] password for astha:
```

```
astha@cdac:~$ id hpc
uid=2026(hpc) gid=2026(hpc) groups=2026(hpc)
astha@cdac:~$
```

2. Add user with comment. (Comment: "High Performance Computing")

```
astha@cdac:~$ tail -l /etc/passwd
nm-openvpn:x:121:122:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
astha:x:1000:1000:astha:/home/astha:/bin/bash
riya:x:1001:1001:riya,1,123456789,123456789,123456789:/home/riya:/bin/bash
muskan:x:1002:1002:,,,:/home/muskan:/bin/bash
cdac:x:1003:1003:,,,:/home/cdac:/bin/bash
punjab:x:1004:1004:,,,:/home/punjab:/bin/bash
chandigarh:x:1005:1005:,,,:/home/chandigarh:/bin/bash
haryana:x:1006:1006:,,,:/home/haryana:/bin/bash
hpc:x:2026:2026::/home/hpc:/bin/sh
WBL:x:2027:2027:High Performance Computing:/home/WBL:/bin/sh
astha@cdac:~$
```

3. Modify GID for the HPC user as 2028.

```
astha@cdac:~$ sudo usermod -u 2028 hpc
astha@cdac:~$ id hpc
uid=2028(hpc) gid=2026(hpc) groups=2026(hpc)
astha@cdac:~$
```

4. Delete HPC user with their directory.

```
astha@cdac:~$ id hpc
uid=2028(hpc) gid=2026(hpc) groups=2026(hpc)
astha@cdac:~$ sudo userdel -r hpc
userdel: hpc mail spool (/var/mail/hpc) not found
userdel: hpc home directory (/home/hpc) not found
astha@cdac:~$ id hpc
id: 'hpc': no such user
astha@cdac:~$
```

5. Create "Programming" user.

```
systemd-oom:x:990:990:systemd Userspace OOM Killer:/:usr/sbin/nologin
kernoops:x:106:65534:Kernel Oops Tracking Daemon,,,:usr/sbin/nologin
whoopsie:x:107:109:/:nonexistent:/bin/false
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:usr/sbin/nologin
avahi:x:108:111:Avahi mDNS daemon,,,:run/avahi-daemon:usr/sbin/nologin
tcpdump:x:109:112:/:nonexistent:usr/sbin/nologin
sssd:x:110:113:SSSD system user,,,:var/lib/sss:usr/sbin/nologin
speech-dispatcher:x:111:29:Speech Dispatcher,,,:run/speech-dispatcher:/bin/false
cups-pk-helper:x:112:114:user for cups-pk-helper service,,,:nonexistent:usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:usr/sbin/nologin
saned:x:113:116:/:var/lib/saned:usr/sbin/nologin
geoclue:x:114:117:/:var/lib/geoclue:usr/sbin/nologin
cups-browsed:x:115:114:/:nonexistent:usr/sbin/nologin
hplip:x:116:7:HPLIP system user,,,:run/hplip:/bin/false
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:usr/sbin/nologin
rtkit:x:117:119:RealtimeKit,,,:proc:usr/sbin/nologin
colord:x:118:120:colord colour management daemon,,,:var/lib/colord:usr/sbin/nologin
gnome-initial-setup:x:119:65534:/:run/gnome-initial-setup:/bin/false
gdm:x:120:121:Gnome Display Manager:/var/lib/gdm3:/bin/false
nm-openvpn:x:121:122:NetworkManager OpenVPN,,,:var/lib/openvpn/chroot:usr/sbin/nologin
astha:x:1000:1000:astha:/home/astha:/bin/bash
riya:x:1001:1001:riya,1,123456789,123456789:/home/riya:/bin/bash
muskan:x:1002:1002:,,,:/home/muskan:/bin/bash
cdac:x:1003:1003:,,,:/home/cdac:/bin/bash
punjab:x:1004:1004:,,,:/home/punjab:/bin/bash
chandigarh:x:1005:1005:,,,:/home/chandigarh:/bin/bash
haryana:x:1006:1006:,,,:/home/haryana:/bin/bash
WBL:x:2027:2027:High Performance Computing:/home/WBL:/bin/sh
Programming:x:2028:2028:/:home/Programming:/bin/sh
astha@cdac:~$
```


6. Assign "Node" group as primary, and "Cluster" group as secondary for the "Programming" user.

```
astha@cdac:~$ tail -l /etc/group
punjab:x:1004:
chandigarh:x:1005:
haryana:x:1006:
mohali:x:1007:
zirakpur:x:1008:
panchkula:x:1009:
WBL:x:2027:
Programing:x:2028:
node:x:2029:
cluster:x:2030:
astha@cdac:~$
```

```
punjab:x:1004:
chandigarh:x:1005:
haryana:x:1006:
mohali:x:1007:
zirakpur:x:1008:
panchkula:x:1009:
WBL:x:2027:
Programing:x:2028:
node:x:2029:
cluster:x:2030:Programing
astha@cdac:~$
```

```
astha@cdac:~$ id Programing
uid=2028(Programing) gid=2029(node) groups=2029(node),2030(cluster)
astha@cdac:~$
```

7. Delete "Programming" user without deleting directory.

```
astha@cdac:~$ id Programing
uid=2028(Programing) gid=2029(node) groups=2029(node),2030(cluster)
astha@cdac:~$ sudo userdel Programing
userdel: group Programing not removed because it is not the primary group of user Programing.
astha@cdac:~$ id Programing
id: 'Programing': no such user
astha@cdac:~$
```